Marine Ecosystem Risk Documentation

Project Overview

The Marine Ecosystem Risk Assessment Dashboard is an interactive Power BI solution designed to analyse environmental factors affecting coral reef ecosystems across major marine regions. The dashboard combines **ocean temperature, pH levels, marine heatwave occurrence, bleaching severity**, and species observations to evaluate ecological risk and identify vulnerable marine locations.

This project aims to demonstrate how environmental indicators can be transformed into meaningful insights through **Data modelling**, **DAX calculations**, and **Interactive visualizations**.

Research Objective

Coral reef ecosystems are increasingly threatened by rising **sea surface temperatures, ocean acidification,** and **marine heatwaves**. Understanding the relationship between these factors is essential for identifying regions at greater ecological risk and supporting conservation efforts.

The dashboard was developed to answer the following questions:

- Which marine locations exhibit the highest environmental risk?
- How do sea surface temperature (SST) and pH levels vary across regions?
- What is the relationship between marine heatwaves and ocean temperature?
- Which locations show signs of greater ecosystem stress?
- How can multiple environmental indicators be combined into a **single risk assessment framework**?

Data Sources

The dataset contains marine environmental observations

- **Dataset Source:** https://www.kaggle.com/datasets/atharvasoundankar/shifting-seas-ocean-climate-and-marine-life-dataset
- **Timeline:** 2015– 2023
- **Data Size: 500**+ rows and 8 columns
- **Domain:** Marine Geology

Additional calendar tables were created to support time-based analysis.

Attribute (Column /Features) Details:

Attribute	Data Type	Description	Example
Date	Date/Text	Observation date of the coral reef monitoring record	31-12-2023
Location	Categorical	Ocean/reef region where data was collected	Maldives, Red Sea
Latitude	Decimal	Geographic latitude of the observation point	20.0407
Longitude	Decimal	Geographic longitude of the observation point	38.4861
SST (°C)	Decimal	Sea Surface Temperature in degrees Celsius	28.08
pH Level	Decimal	Ocean water acidity/alkalinity measurement	8.098
Bleaching Severity	Categorical	Coral bleaching intensity level	Low, Medium, High
Species Observed	Integer	Number of marine species observed at the location	129
Marine Heatwave	Boolean	Indicates whether a marine heatwave event occurred	True / False

Data Preparation

Data was prepared using:

Excel

- Initial exploration
- Hemisphere classification
- Data validation

Power Query

- Data cleaning
- Data transformation
- Column standardization
- Data type conversion

Data Modelling

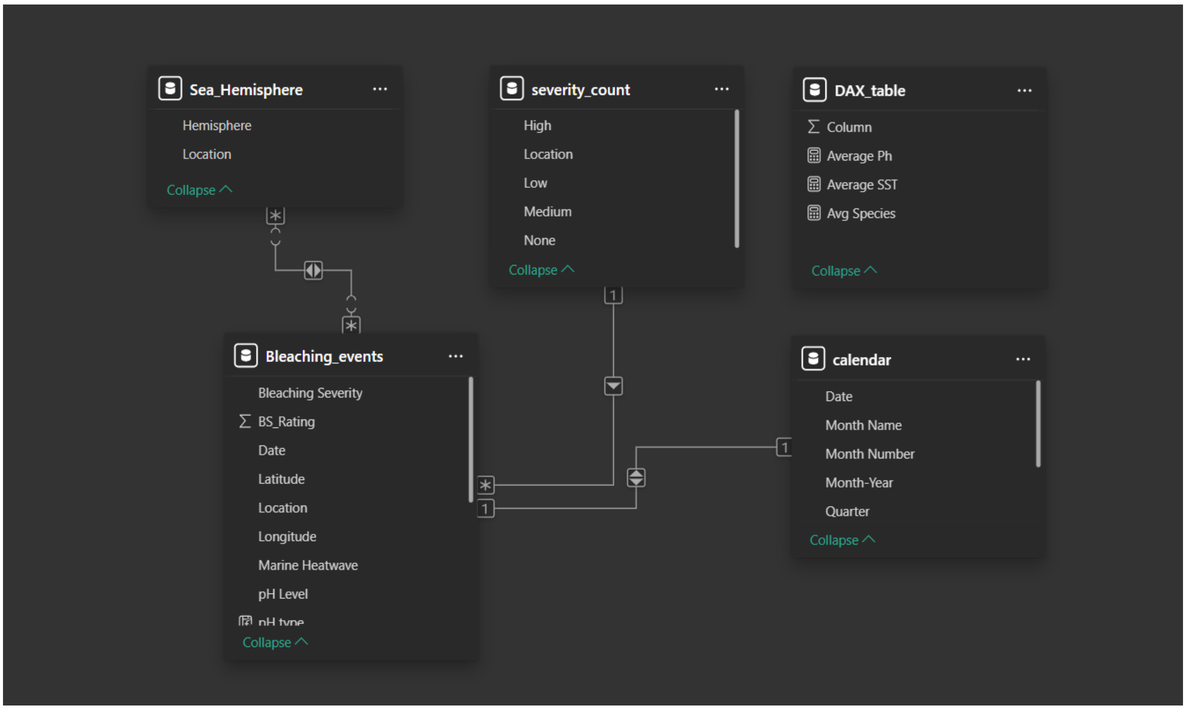
- Relationship creation
- Calendar table implementation
- Time intelligence support

Data Modelling and DAX (Power BI)

- **Data Model:** Established relationships between tables, defined cardinality, and created lookup tables where necessary.

  Bleaching Events: Fact Table
  Sea Hemisphere: Dimension Table
  Severity Count : Dimension Table
  Calendar : Dimension Table

  The schema of this Data modelling is **Star Schema.**
- **Cardinality Obtained:** One to One, Many to One, Many to Many



Calculated Columns & DAX Measures: Implemented DAX formulas for key metrics, such as total sales, growth percentages, and cumulative totals

A Custom Risk Rating Analogy was created for each location by assigning scored for each parameters with the help of Dax

DAX Measures created

- Average Ph
- Average Sea Surface Temperature
- Average Species Observed
- Average risk rating
- Average Bleaching Events count

Key Metrics

The dashboard tracks:

- Average Sea Surface Temperature (SST)
- Average Ocean pH
- Average Species Observed
- Coral Bleaching Events
- Composite Risk Rating
- Location-Based Risk Scores

Analysis and Visualizations

Dashboard Features:

Average pH Card
Displays the average ocean pH level, helping monitor ocean acidity and its impact on marine ecosystems.

Average SST Card
Shows the average Sea Surface Temperature (SST), a key factor influencing coral bleaching and ecosystem stress.

Average Species Card
Represents the average number of species observed, indicating the overall biodiversity of marine habitats.

Year Slicer
Allows users to filter the entire dashboard by year for temporal analysis and comparison.

Location Slicer
Enables filtering of dashboard visuals by specific marine locations to support regional analysis.

Bubble Map
Visualizes the geographical distribution of reef locations using latitude and longitude coordinates.

Species Observed per Year Card
Displays the average number of species observed for the selected filters, helping track biodiversity trends.

Bleaching Events per Year Card
Shows the total number of coral bleaching events, indicating the level of environmental stress on reefs.

Average Risk Rating Card
Provides an overall risk score summarizing marine ecosystem vulnerability across selected locations.

Risk Rating by Location Table
Presents location-wise average risk ratings, making it easy to compare ecosystem risk levels across regions.

Ocean Parameters Combo Chart
Compares average pH, risk rating, and species count across locations to identify relationships between environmental conditions and biodiversity.

Average SST & pH by Location Line Chart
Shows how sea surface temperature and pH levels vary across locations, helping identify areas facing environmental stress.

Analytical Approach

The dashboard follows a risk-assessment framework:

Ocean Conditions
(SST+ pH)
↓
Environmental Stress
(Marine Heatwaves)
↓
Ecosystem Impact
(Bleaching + Species Observed)
↓
Risk Assessment
(Risk Rating)

A custom risk scoring methodology was developed using environmental indicators to compare ecosystem vulnerability across locations.

1. Descriptive Analysis

Findings

- Average SST is approximately **29°C**.
- Average ocean pH is approximately **8.05**.
- Average species observed is approximately **120**.
- Multiple locations experience marine heatwave events.
- Risk ratings vary across locations, with some regions showing significantly higher ecosystem risk

2. Diagnostic Analysis

Findings

- Locations with higher SST generally exhibit lower pH values.
- Marine heatwave occurrences are associated with elevated sea surface temperatures.
- Areas experiencing environmental stress tend to have higher risk scores.
- Variations in species observations suggest differing ecosystem resilience across locations.

3. Predictive Analysis

Predictions

- Locations with consistently high SST are likely to experience greater environmental stress.
- Continued increases in SST may further reduce pH levels through ocean acidification processes.
- Regions already displaying elevated risk scores may be more susceptible to future bleaching events.
- Biodiversity may decline in locations where environmental conditions continue to deteriorate.

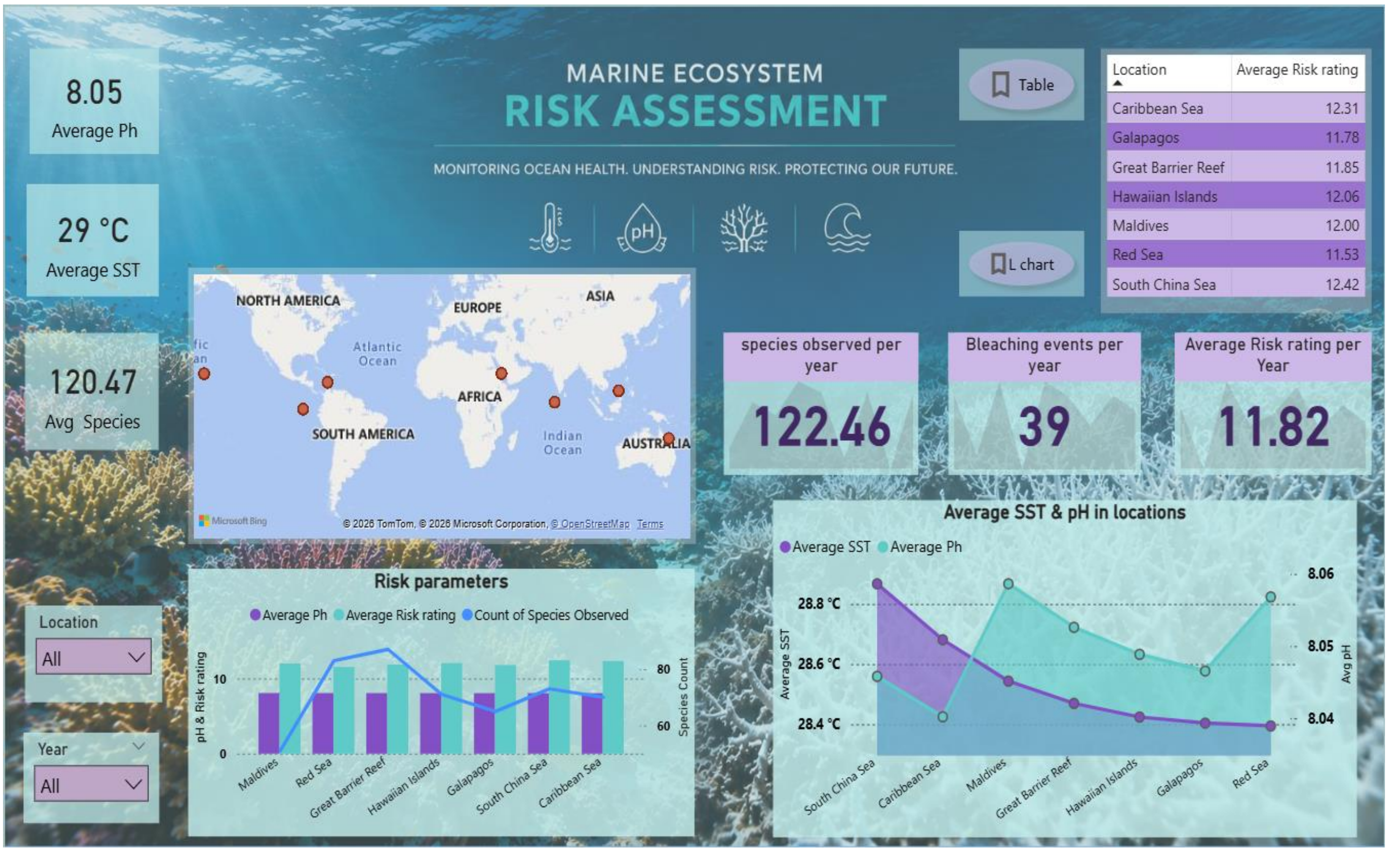
4. Prescriptive Analysis

Recommendations

High-Risk Locations

- Prioritize conservation efforts in locations with the highest risk ratings.

- Marine Heatwave Monitoring
- Implement early warning systems for regions experiencing frequent heatwave events.
- Temperature Management
- Support climate mitigation strategies aimed at reducing ocean warming.
- Coral Protection Programs
- Increase restoration and monitoring initiatives in vulnerable reef systems.
- Biodiversity Monitoring
- Conduct regular ecological assessments in regions showing declining species observations.



Key Insights

- **Higher SST levels increase** the likelihood of **coral bleaching**.
- Prolonged exposure **to elevated temperatures** can weaken coral resilience and reduce **ecosystem stability**.
- Temperature emerges as one of the strongest environmental risk drivers in this dataset.
- Although current **pH levels** remain within a normal marine range, slight decrease in pH can **significantly affect coral growth and reef formation**.
- Continued ocean acidification could reduce the ability of corals to build calcium carbonate skeletons.
- Maintaining stable pH levels is critical for long-term ecosystem sustainability. Higher species counts generally indicate healthier and more resilient ecosystems.
- Some locations exhibit lower biodiversity despite favourable environmental conditions, suggesting additional ecological pressures.
- **Biodiversity** should be protected as it contributes directly to **ecosystem stability** and **recovery from environmental disturbances.**
- Bleaching events are strongly associated with elevated sea temperatures.
- Frequent bleaching can reduce coral cover, disrupt marine food chains, and threaten biodiversity.
- The observed bleaching frequency highlights the need for proactive conservation measures

- **Higher sea surface temperatures** cause environmental stress from bleaching activity causes increased **ecosystem vulnerability** compared to other regions.
- These locations should be prioritized for monitoring and conservation efforts
- The **Red Sea** records the **lowest average risk rating** of 11.53. This Indicates comparatively better ecosystem conditions, Suggests greater resilience to current environmental stressors.
- Red sea could serve as a benchmark for studying factors that contribute to ecosystem stability.
- As **SST increases**, ecosystem risk generally increases. This relationship supports existing scientific evidence linking ocean warming to **coral bleaching** and **habitat degradation**

Project Outcome:

The dashboard reveals that marine ecosystems are experiencing moderate to high environmental risk due to elevated sea surface temperatures, ongoing coral bleaching events, and ecological stress across all monitored locations**.** The **South China Sea** and **Caribbean Sea** emerge as the most **vulnerable regions**, while the **Red Sea** demonstrates relatively **lower risk.**

Although biodiversity remains reasonably healthy and ocean pH is currently stable, continued warming trends and bleaching activity suggest that **proactive conservation**, **temperature monitoring**, and **reef protection strategies** are essential to ensure long-term **marine ecosystem sustainability.**
